IIA • WEB DESIGN

UNIT 03

Introduction to HTML

Create your first information webpage

LIBRARY & INFORMATION STUDIES • RECORDS MANAGEMENT SYSTEMS

LEARNING GOALS

By the end of this unit…

Understand HTML

Explain markup, elements and the roles of HTML, CSS and JavaScript.

Recognise a complete HTML file.

Build and test

Write headings and paragraphs.

Save index.html, open it in a browser and fix basic mistakes.

CONTEXT

Why HTML matters to information professionals

Library services

Opening hours, borrowing guidance and contact details can be published as structured HTML.

Records centres

Public procedures and authorised request guidance become understandable, accessible pages.

Shared foundation

Both programmes use the same HTML syntax; professional needs determine the content.

CONCEPTS

The three main web technologies

HTML = meaning

Defines headings, paragraphs, lists and other content elements.

CSS = presentation

Controls colour, layout, font choices and spacing (Unit 7).

JavaScript = behaviour

Can respond to clicks and update page content (Unit 10).

DEFINITION

HTML is a markup language

HyperText Markup Language

HTML uses tags to identify the role of webpage content. A browser reads this structure and displays the page. HTML alone does not securely store records or manage library accounts.

A useful first webpage does not need a server or database.

DOCUMENT STRUCTURE

A complete first HTML file

HTML SOURCE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Library Services</title>
</head>
<body>
  <h1>Library Services</h1>
  <p>Welcome to our library.</p>
</body>
</html>
```

EXPLANATION

The head contains information about the page.

The body contains the heading and paragraph visitors can read.

DOCUMENT STRUCTURE

The declaration and the root element

DOCTYPE

<!DOCTYPE html> tells browsers to use modern HTML rendering.

It is a declaration, not an ordinary HTML tag.

HTML root

<html lang="en"> contains the document.

The lang attribute tells software that the primary language is English.

DOCUMENT STRUCTURE

HEAD versus BODY

HEAD: about the page

Includes metadata, the browser-tab title and links to stylesheets.

Most head information is not displayed as page text.

BODY: visible content

Contains paragraphs, headings, later images, links, tables and forms.

The browser renders these for visitors.

DOCUMENT STRUCTURE

Why metadata matters

Charset

<meta charset="UTF-8"> helps display characters and symbols correctly.

Viewport

A viewport meta tag helps layouts fit the width of mobile screens.

Page title

<title> names a browser tab; it differs from the on-page <h1> heading.

SYNTAX

Opening tag + content + closing tag

HTML SOURCE

```
<p>Library hours are available.</p>

<h1>Records Centre</h1>
```

EXPLANATION

 begins a paragraph.

The text inside is its content.

 ends the paragraph with a slash.

The entire structure is one HTML element.

SYNTAX

Correctly nested elements

HTML SOURCE

```
<!-- Correct -->
<p>Ask a <strong>librarian</strong>.</p>

<!-- Incorrect -->
<p>Ask a <strong>librarian</p></strong>
```

EXPLANATION

An inner element must finish before its outer element finishes.

Well-formed nesting helps browsers and people interpret the page.

SYNTAX

Three terms to distinguish

Tag

<h1> and </h1> are HTML tags that mark a heading.

Element

<h1>Our Library</h1> is a complete element.

Attribute

lang="en" adds information to an opening tag.

ATTRIBUTES

Attributes add meaning or identifiers

HTML SOURCE

```
<html lang="en">

<p id="hours" title="Service hours">
  We open at 8 a.m.
</p>
```

EXPLANATION

Attributes belong in an opening tag.

The name and quoted value explain a property of the element.

IDs should be unique within a page.

ATTRIBUTES

Common attributes in Unit 3

lang

Communicates the document language to software and assistive technologies.

id

Identifies one element uniquely for linking, styling or scripting.

class

Groups multiple elements for later shared styling or behaviour.

TEXT STRUCTURE

Headings organise content

HTML SOURCE

```
<h1>Community Library</h1>
<h2>Borrowing Services</h2>
<h3>Loan Periods</h3>
<p>Ask staff for borrowing rules.</p>
```

EXPLANATION

<h1> is the main heading.

<h2> introduces a subsection.

<h3> belongs under that subsection.

Use levels for structure, not merely font size.

TEXT STRUCTURE

Paragraphs are separate ideas

Use <p> to group sentences

A paragraph is a distinct block of text. Multiple spaces and line breaks in ordinary HTML source do not normally produce extra visible spacing. Create a new paragraph with another <p> element.

Keep paragraphs clear and short for students and service visitors.

TEXT STRUCTURE

Line breaks and thematic breaks

Starts a line break within one block when a break is meaningful, such as an address.

It is a void element.

<hr>

Indicates a thematic shift between sections.

It is not a tool for adding lots of blank space.

TEXT MEANING

Strong importance and emphasis

HTML SOURCE

```
<p><strong>Important:</strong>
  Bring your library card.</p>

<p><em>Authorised users</em>
  may request restricted records.</p>
```

EXPLANATION

 communicates importance.

 communicates stress emphasis.

Browsers commonly show bold and italic styles, but meaning comes first.

SPECIAL CHARACTERS

How to display reserved characters

HTML SOURCE

```
<p>Research &amp;amp; Learning</p>

<p>Write &amp;lt;p&amp;gt; to show
  a paragraph tag as text.</p>
```

EXPLANATION

Use & for an ampersand.

Use < for < and > for >.

These character references prevent the browser from reading displayed symbols as markup.

SOURCE CODE

Comments and privacy

HTML comment

<!-- Notes for future editors -->

Comments explain source code and normally do not appear on the visible page.

Never store secrets

Anyone can view page source. Do not insert passwords, private borrower information or protected record IDs in comments.

BUILDING LOCALLY

Five steps to your first webpage

1. Edit

Open Notepad or another plain-text editor and write the HTML.

2. Save

Save as index.html, not index.html.txt. UTF-8 is a sensible encoding.

3. Open

Double-click the file to open it in a browser; after edits, save and refresh.

FILES

Why the name index.html is useful

Recognise a real HTML file

An .html extension tells the operating system and browser that the file contains HTML. The name index.html is commonly used for the main page of a site or folder. A local file can open at a file:/// address.

If tags appear as visible text, check the extension and the file you opened.

DEBUGGING

Three common beginner problems

Content does not update

You changed the source but forgot to save. Save the file and refresh the browser tab.

Page structure looks wrong

Check closing tags, nesting and whether visible content was placed inside <body>. Use View Page Source.

APPLIED EXAMPLE

Library information page

HTML SOURCE

```
<h1>Community Library</h1>
<h2>Our Services</h2>
<p>We help readers find books.</p>
<p><strong>Hours:</strong>
 Monday to Friday.</p>
```

EXPLANATION

A heading identifies the institution.

A subsection heading names the service.

Paragraphs provide useful public information.

No real borrower data is included.

APPLIED EXAMPLE

Records-centre information page

HTML SOURCE

```
<h1>Records Centre</h1>
<h2>Request Guidance</h2>
<p>Ask staff about authorised access.</p>
<p><strong>Hours:</strong>
 Weekdays, 9 a.m. to 4 p.m.</p>
```

EXPLANATION

The HTML syntax matches the library example.

The text suits records-management users.

Restricted data must stay out of public HTML pages.

PRACTICAL WORK

Build a fictional information page

Prepare

Choose a library or records centre. Enter a title and a main heading.

Write

Add two section headings and three short paragraphs about services.

Test

Save index.html, open it in a browser and check tab title and heading order.

PRACTICE TOOL

Live HTML laboratory on the website

Edit source, then update the preview

The interactive editor on the Unit 3 page contains library, records-centre and basic HTML examples. Change a heading or paragraph, preview the result, then download index.html to test it locally.

The preview runs without uploading your code to any server.

QUICK REVIEW

Check the roles of the main parts

DOCTYPE

Modern HTML declaration. Not an ordinary element.

HEAD

Page metadata and browser-tab title.

BODY

The visible webpage content and structure.

ASSESSMENT

Simple assignment and self-check quiz

Individual assignment

Make one index.html page with a title, one h1, two h2 headings and three paragraphs. Submit the file plus a screenshot when instructed.

Quiz

Eight questions are selected from a larger bank and shuffled. Submit to see your score and corrections for missed questions.

REVIEW

Six important points to remember

Content first

HTML represents meaning; CSS and JavaScript have different roles.

Structure matters

Use clear nesting, headings and paragraphs.

Test the file

Save .html correctly, refresh often and use fictional data.

WHAT IS NEXT?

Unit 4: Organising Web Content

From a page to a small website

Next, you will learn links, lists, images and file paths. These help visitors move between related pages and find information efficiently. Later units add tables, forms and visual styling.

Independent revision: MDN HTML and the W3Schools HTML tutorial.